DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 4

Machine learning engineer

Profession 4 of 10 in Development and IT.

What the profession means

A machine learning engineer builds systems that learn patterns from data and make predictions: lead scores, demand forecasts, fraud flags. The work covers data preparation, model training and testing, deployment and monitoring.

What professionals do every day

Cleans and prepares data, selects features, trains and compares models, measures them honestly on data they have not seen, deploys the chosen model and watches it for drift.

Why businesses hire them, and what problems they solve

Many business decisions depend on predictions: who will buy, what will sell, which invoice is risky. An engineer turns historical data into a model the business can use, and says how much to trust it.

- Guessing which leads to call first.
- Stock or demand planning based on gut feel.
- Manual checks on large volumes of transactions.

Skills you need

- Python, NumPy and pandas.
- Statistics: averages, spread, correlation, the idea of a test set.
- Scikit-learn for classical models.
- Evaluation: accuracy, precision, recall and the cost of each error.
- Deployment basics and monitoring.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Python	Language for data and ML	Free	Free	www.python.org
scikit-learn	Classical ML models	Free	Free	scikit-learn.org
pandas	Data tables	Free	Free	pandas.pydata.org
Jupyter	Notebooks for experiments	Free	Free	jupyter.org
Kaggle	Datasets and practice competitions	Free	Paid features	www.kaggle.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	Clean a dataset and train one classifier with a test set.
Intermediate	Compare three models and explain the choice.

Advanced	Deploy a model behind an API and log its predictions.

Projects by level

- **Beginner:** Lead scoring for a sales team.
- **Intermediate:** Churn prediction with a retention plan.
- **Advanced:** Demand forecast with error bands.

Client-ready portfolio project

A lead-scoring model for a fictional education company: data preparation, a model, an honest test score, and a one-page note on what the score can and cannot tell the sales team.

Worked example

Worked example. Time on the pricing page and emails opened predict conversion. The model is trained on 75% of the data and scored on the remaining 25%, which it has never seen.

Assignment with full solution

ASSIGNMENT

A lead-scoring model with an honest test score

Business problem: A sales team wants to call the most likely buyers first. The model must be tested on data it did not learn from.

Steps

1. Load the data and check for missing values.
2. Split into training and test sets with a fixed random seed.
3. Train a logistic regression model.
4. Report accuracy on the test set and state the expected error cost (for example a missed buyer).
5. Write one paragraph on what the score does not show.

Expected output: A test accuracy above the agreed threshold (here 0.8) and a written limitation note.

SOLUTION (ANSWER KEY)

The test set is held back during training, so the score reflects new data. A fixed `random_state` makes the result repeatable. Accuracy alone can hide problems when one outcome is rare; check precision and recall in a real project. The test here passes at about 0.86 accuracy on synthetic data.

PYTHON

```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

rng = np.random.default_rng(42)
n = 400
time_on_site = rng.uniform(0, 10, n)          # minutes on the pricing page
```

```
emails_opened = rng.integers(0, 8, n)         # emails opened this month
noise = rng.normal(0, 1.0, n)
score = 0.6 * time_on_site + 0.5 * emails_opened + noise
converted = (score > 5.5).astype(int)         # 1 = became a paying customer

X = np.column_stack([time_on_site, emails_opened])
X_train, X_test, y_train, y_test = train_test_split(X, converted, test_size=0.25,
random_state=0)
model = LogisticRegression().fit(X_train, y_train)
acc = accuracy_score(y_test, model.predict(X_test))
if __name__ == "__main__":
    print("test accuracy:", round(acc, 3))
    assert acc > 0.8, acc
    print("lead model test passed")
```

Common mistakes

- Testing on the training data, which makes results look better than they are.
- Using a model without checking for data leakage (information from the future).
- Reporting only accuracy when classes are unbalanced.
- Not monitoring a deployed model as customer behaviour changes.

Freelancing opportunities

- Lead-scoring and churn models for small firms.
- Forecasting for inventory or sales.
- Freelance briefs for 'machine learning', 'scikit-learn', 'predictive model'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- scikit-learn user guide
- pandas documentation
- Kaggle Learn (free micro-courses)

Part of **Master Freelancing: AI and Automation**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.